25901556 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCSAC200 - HEART RATE TRAINING

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. The principle of overload means that exercise training should [U] / [CO5]

   a). Be enjoyable at all times b). Gradually increase the training demand

   c). Focus only on flexibility d). Avoid fatigue

2. During endurance training, stroke volume increases mainly due to [U] / [CO3]

   a). Increased ventricular filling b). Increased heart rate

   c). Decreased blood volume d). Reduced cardiac output

3. An athlete exercising in a hot environment shows a higher heart rate mainly because [U] / [CO3]

   a). Stroke volume decreases b). Oxygen consumption decreases

   c). Muscle contraction reduces d). Blood is redirected to the skin for cooling

4. Chronic heart rate monitoring is mainly used to [U] / [CO5]

a). Measure instant fatigue

b). Assess long-term adaptation to training

c). Detect immediate overexertion

d). Calculate target HR zones

5. The main function of the heart is to [U] / [CO1]

a). Filter blood

b). Produce hormones

c). Pump blood

d). Store oxygen

6. The human heart is located in the [U] / [CO1]

a). Abdominal cavity

b). Mediastinum

c). Right lung

d). Neck

7. A patient has normal systolic pressure but elevated diastolic pressure. This condition most likely indicates increased [An] / [CO3]

a). Cardiac output

b). Blood volume

c). Peripheral resistance

d). Heart rate

8. Resting heart rate is generally higher in children than in adults mainly because children have [U] / [CO3]

a). Higher metabolic rate

b). Larger heart size

c). Greater blood volume

d). Stronger cardiac muscles

9. The main purpose of a warm-up before running is to [U] / [CO6]

a). Reduce body temperature

b). Increase fatigue

c). Improve flexibility only

d). Prepare muscles and cardiovascular system for activity

10. Which method provides the most accurate determination of maximum heart rate? [U] / [CO3]

a). Age-predicted formula

b). Resting heart rate calculation

c). Submaximal exercise test

d). Maximal graded exercise test

[1x10 = 10]

Part B

Short Answer

Answer any **10** questions

Each question carries **2** marks

11. Define afterload? [K] / [CO2]

12. Explain how an increase in vascular resistance affects blood pressure. [U] / [CO3]

13. State the relationship between heart rate and oxygen consumption during exercise. [U] / [CO7]

14. Explain what is meant by walking training zones and state their significance in exercise prescription. [U] / [CO6]

15. Explain the importance of ventricular diastole in the cardiac cycle. [U] / [CO2]

16. How does increased afterload influence stroke volume? [U] / [CO2]

17. Calculate the maximum heart rate of a 20-year-old using the standard formula. [A] / [CO3]

18. Differentiate between continuous training and interval training for endurance development. [An] / [CO5]

19. How does training in the anaerobic zone differ from training in the aerobic zone? [U] / [CO6]

20. Explain the role of venous valves in venous return. [U] / [CO2]

21. Why is plasma volume important for cardiovascular efficiency during exercise? [U] / [CO2]

22. Define heart rate training zone. [K] / [CO3]

[2x10 = 20]

Part C

Short Essay

Answer any **4** questions

Each question carries **5** marks

23. Define energy expenditure and explain its relationship with heart rate during exercise. [U] / [CO4]

24. Explain how electrical impulses are generated and conducted through the heart to produce coordinated cardiac contraction. [U] / [CO1]

25. Explain the methods used to measure VO_2 max and analyze its importance in assessing cardiovascular fitness. [An] / [CO3]

26. Describe the factors affecting ventricular filling time. [U] / [CO2]

27. Define cardiac output. Explain the formula used to calculate cardiac output and describe its normal resting values. [U] / [CO3]

28. Analyze how endurance training improves aerobic performance by affecting oxygen uptake and utilization. [An] / [CO4]

[5x4 = 20]